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Project Title	High-Performance Market Basket Analytics with Compressed FP-Trees from Scratch
Week	Week 4

WEEK-4 TASK

Task Name

High-Performance Market Basket Analytics with Compressed FP-Trees from Scratch

1. Introduction

Market Basket Analysis finds items that are frequently purchased together. FP-Growth uses a compressed **FP-Tree** to efficiently discover these patterns.

2. Objectives

- Find frequent item sets.
- Build a compressed FP-Tree.
- Use node-link pointers.
- Construct conditional FP-Trees.
- Generate high-confidence association rules.
- Improve processing efficiency.

3. Market Basket Analysis

It identifies relationships between products in transactions.

Example: **Bread + Milk → Butter**

4. FP-Tree Structure

An FP-Tree stores transactions in a compact tree structure. Each node contains:

- Item name
- Count
- Parent pointer
- Child pointers
- Node-link pointer

5. FP-Tree Construction

1. Count item frequencies.
2. Remove infrequent items.
3. Sort frequent items.
4. Insert transactions into the tree.
5. Share common prefixes.

6. Node-Link Pointers

Node links connect nodes containing the same item. They help quickly find all occurrences of an item in the FP-Tree.

7. Conditional Pattern Base

It contains the prefix paths related to a selected frequent item. These paths are used to find additional frequent patterns.

8. Conditional FP-Tree

A smaller FP-Tree is created from the conditional pattern base. It is mined recursively to find frequent item sets.

9. Frequent Pattern Mining

The algorithm:

1. Builds the FP-Tree.
2. Selects frequent items.
3. Creates conditional pattern bases.
4. Builds conditional FP-Trees.

5. Extracts frequent itemsets.

10. Association Rules

Association rules show relationships between items.

Example:

Bread → Milk

Confidence = $\text{Support}(\text{Bread, Milk}) / \text{Support}(\text{Bread})$

11. High-Confidence Rules

Rules with confidence above the minimum threshold are selected as strong association rules.

12. Performance Optimization

Performance is improved using:

- Tree compression
- Shared prefixes
- Node links
- Removal of infrequent items
- Recursive mining
- Reduced candidate generation

13. Applications

- Product recommendations
- Retail analysis
- Cross-selling
- Product placement
- Customer behavior analysis
- Personalized marketing

14. Expected Outcome

The system should efficiently discover frequent itemsets and generate strong association rules from large transaction datasets.

15. Conclusion

Compressed FP-Trees provide an efficient method for Market Basket Analysis. They reduce redundant storage and help discover useful product relationships quickly.